Question	Answer	Marks	AO	Guidance
8	$\frac{dy}{dx} = 6x^2 + 6mx - 9m$	**M1***	**3.1a**	differentiation of all 4 terms with 3 of the 4 terms differentiated correctly
		A1	**1.1**	all correct
	$(6m)^2 - 4 \times 6 \times (-9m)$ **oe** seen	**M1dep***	**3.1a**	discriminant for their $6, 6m$ and $-9m$; may see $(2m)^2 - 4 \times 2 \times (-3m)$
	two values of m obtained from their discriminant $36m^2 + 216m < 0$ or $36m^2 + 216m = 0$ **oe**	**M1**	**2.1**	dependent on obtaining discriminant from their derivative; **M0** for use of their discriminant > 0 or ≥ 0 **NB** $4m^2 + 24m < 0$ or $4m^2 + 24m = 0$
	0 and – 6 identified	**A1**	**1.1**	
	$-6 < m < 0$ **oe**	**A1**	**3.2a**	inequality or interval must be strict
		[6]		